


UNIVERSITY OF SALERNO

Department of Computer Science

Bachelor's Degree in Computer Science

BACHELOR'S THESIS

# StARAPTOR: Empowering Player Voices in Game Development Pipelines via LLM-Powered Feature Extraction, Association Mining, and Player Count Forecasting

SUPERVISOR

Prof. Fabio Palomba

University of Salerno

CANDIDATE

**Alfonso Maria Vicidomini**

Student ID: 0512113682

*This thesis has been realized in*



*Dedica o citazione*

**Abstract**

Game Software Engineering (GSE) differs from traditional enterprise software because player preferences concerning narrative, mechanics, or balance are abstract and difficult to quantify, especially prior to release. While the rise of live-service games and early-access models has shifted production toward continuous iteration, how informal player feedback can be systematically converted into formal requirements remains poorly defined.

To address this gap, this thesis proposes StARAPTOR (a Steam Automated Review Association Miner for Player-count Time-series Optimized Regression), a prototype tool that leverages Large Language Models (LLMs) to extract discrete attributes from Steam reviews. The tool expands feature dimensionality using FP-Growth association mining to identify recurring relationships, combining them into a feature matrix that feeds an XGBoost time-series regressor to forecast daily mean concurrent player counts. By mathematically linking player sentiment to retention-related metrics, StARAPTOR automates feedback analysis and helps developers quantify the impact of user requirements on player engagement.

# Contents

**List of Figures** ii

**List of Tables** iii

**1 Introduction** 1

1.1 Background . . . . . 1

1.2 Application Context . . . . . 2

1.2.1 State-of-the-Art Brief . . . . . 2

1.2.2 State-of-the-Practice Brief . . . . . 2

1.3 Objectives . . . . . 4

**2 Esempio Capitolo** 5

2.1 Prima sezione . . . . . 5

2.1.1 Prima sottosezione . . . . . 5

**3 Conclusions and Future work** 10

**Bibliography** 11

# List of Figures

2.1 Girasole piantato nel cortile del Dipartimento di Informatica. . . . . 7

# List of Tables

2.1 Questions to decide whether to conduct a MLR [**?**]. . . . . 6
2.2 Esempio di tabella . . . . . 7

CHAPTER 1

# Introduction

## 1.1 Background

**Game Development and Engineering Standards** Game development is a branch of software engineering focused on producing video games. Because the industry expanded relatively late in the West (growing significantly only after the 1985 US release of the Nintendo Entertainment System), its production pipelines lack the formal definitions common in enterprise software development. Whereas organizations such as ISO, IEC, and IEEE publish globally accepted standards for software life cycle processes [1], game development relies on less formalized workflows. McKenzie et al. note similarities with Agile methodologies, but game development does not strictly adhere to them [2].

**Evolution of Game Development and Distribution** Over the past two decades, the video game industry has undergone one of the most significant paradigm shifts in software. Before the widespread adoption of high-speed internet, games were distributed "as-is" with no assurance of maintenance or evolution. Software updates were scarce, and physical "game expansions" incurred prohibitive distribution costs, restricting them to high-budget projects. Consequently, maintenance and evolution

were not central to the software life cycle. Today, continuous updates and downloadable content (DLC) are the norm. The traditional release model is increasingly being superseded by Games-as-a-Service (GaaS) and Early Access (EA) release strategies, which introduce fundamentally different development pipelines and player behaviors [3, 4].

**Impact on Development Pipelines** This shift has forced production pipelines to become faster and more iterative. However, little work has been done to formalize requirements extraction and user feedback validation. Meanwhile, the player base and the volume of feedback have grown exponentially, driven by integrated review systems on digital distribution platforms. This often creates a critical gap between user needs and developer deliverables. Furthermore, researchers have yet to reach a consensus on the optimal approach to aggregate and analyze the vast amount of unstructured feedback available online. Addressing this gap is especially critical in modern development environments, where prevailing production models increasingly prioritize player feedback [4, 5].

## 1.2 Application Context

### 1.2.1 State-of-the-Art Brief

Rajapakshe (2019) noted that no tools existed to fully automate the processing of feedback, leaving the development of such tools as future work [5]. However, recent advances in artificial intelligence (AI) have substantially simplified feature extraction and sentiment analysis. It is now possible to conduct reliable aspect-based classification on data that were previously limited to binary classification [6]. The broader state of the art is discussed in detail in Chapter **??**.

### 1.2.2 State-of-the-Practice Brief

Most off-the-shelf (OTS) game engines provide built-in trackers for player metrics. These metrics can be integrated into business intelligence (BI) pipelines using OTS

tools (e.g., Tableau[1]) or custom machine learning pipelines. Distribution platforms such as Steam also provide developer APIs to collect Key Performance Indicators (KPIs). Recently, third-party analytics platforms such as GameAnalytics[2] and devtodev[3] have gained popularity.

These tools offer default analytics pipelines and increasingly allow custom pipeline generation via natural language prompts. An AI agent interprets the prompt and builds the necessary pipeline by interfacing directly with the provided APIs and SDKs. Although versatile, this approach is vulnerable to hallucinations from large language models (LLMs), leaving users with little control over the pipeline's logic or the correctness of its output, because current architectures can only mitigate rather than eliminate the problem [7]. Consequently, these plug-and-play solutions are often preferred by small independent teams because of lower costs, despite the inherent risks.

Crucially, while these solutions provide deep insights into player metrics and behavior, they largely overlook qualitative player feedback. This makes it easy to model what users do, but not why they do it. This limitation is particularly problematic when addressing ethical concerns in game development, where textual reviews and community posts are a crucial source of information [8]. Translating natural language into structured data remains a complex challenge. As discussed in Chapter **??** and Section **??**, traditional natural language processing (NLP) algorithms face significant limitations, particularly with game reviews, which are frequently noisy, highly contextual, or sarcastic.

In the adjacent field of mobile applications, Apple has described a pipeline of small, fine-tuned LLMs and embedding models that extract features from App Store reviews[4]. Although Google has not published the architecture used for the Play Store, it is reasonable to infer that similar technologies are deployed. These platforms also distribute mobile games, yet no comparable, publicly documented solution has been adopted by PC or console distribution platforms, and it remains unclear whether such techniques transfer to game reviews, whose feature space differs

[1] `https://www.tableau.com`

[2] `https://www.gameanalytics.com`

[3] `https://www.devtodev.com`

[4] `https://machinelearning.apple.com/research/app-store-review`

substantially from that of mobile applications. This thesis explores whether an LLM-based extraction pipeline can be adapted to this domain and used in development requirement analysis.

## 1.3 Objectives

**Our Goal.** To design, implement, and evaluate StARAPTOR, a prototype system that converts unstructured player feedback from Steam reviews into quantified inputs for ML-based forecasting of player engagement.

The system addresses three sub-objectives.

- **Extraction of structured attributes from noisy review bodies.** Player reviews are unstructured but information-rich, and manual reading is not feasible with high review throughtput. StARAPTOR uses an LLM to separate noise, such as sarcasm and unrelated content, from signal; this produces an ordinal rating for each design pillar in each review and finding associated qualitative data.
- **Identification of recurring patterns across reviews.** StARAPTOR applies association mining to the extracted attributes to find and understand common patterns. Every mined rule remains traceable to the reviews that support it, because each attribute is tied to the source.
- **Per-game forecasting of engagement metrics.** Different games respond differently to the same design issues, so a single global model would average away the signal. StARAPTOR trains and tunes an XGBoost regressor per game, using the extracted attributes and mined rules as features, to forecast daily average concurrent players and estimate which features are most strongly associated with engagement and retention.

These sub-objectives are addressed by the research questions and evaluated as described in Chapter **??**.

**RQ$_1$.** *Are there any advantages into implementing a*

CHAPTER 2

# Esempio Capitolo

## 2.1 Prima sezione

### 2.1.1 Prima sottosezione

**Primo paragrafo** Questa è una prova di un testo di un paragrafo.

The well known Pythagorean theorem $x^2 + y^2 = z^{21}$ was proved to be invalid ***for other exponents***.

COM.JAVA.XXX

"Lorem ipsum"

Meaning the next equation has no integer solutions:

$$x_m^n + y^n = z^n$$

The well known Pythagorean theorem $x^2 + y^2 = z^2$ was proved to be invalid for other exponents. Meaning the next equation has no integer solutions:

[1] `http://google.com`

**Table 2.1:** Questions to decide whether to conduct a MLR [**?**].

| # | Question | Answer |
|---|---|---|
| 1 | Is the subject "complex" and not solvable by considering only the formal literature? | Yes |
| 2 | Is there a lack of volume or quality of evidence, or a lack of consensus of outcome measurement in the formal literature? | No |
| 3 | Is the contextual information important to the subject under study? | Yes |
| 4 | Is it the goal to validate or corroborate scientific outcomes with practical experiences? | Yes |
| 5 | Is it the goal to challenge assumptions or falsify results from practice using academic research or vice versa? | Yes |
| 6 | Would a synthesis of insights and evidence from the industrial and academic community be useful to one or even both communities? | Yes |
| 7 | Is there a large volume of practitioner sources indicating high practitioner interest in a topic? | Yes |

***Note**: The possible answers to each question are "Yes" or "No". One or more "Yes" responses suggest that it could be useful to conduct a MLR.*



**Figure 2.1:** Girasole piantato nel cortile del Dipartimento di Informatica.

| Col1 | Col2 | Col3 |
|---|---|---|
| 1 | 6 | 787 |
| 2 | 7 | 5415 |
| 3 | 545 | 7507 |
| 4 | 545 | 7560 |
| 5 | 88 | 6344 |

**Table 2.2:** Esempio di tabella

$$x_m^n + y^n = z^n$$

```
import numpy as np

def incmatrix(genl1,genl2):
    m = len(genl1)
    n = len(genl2)
    M = None #to become the incidence matrix
    VT = np.zeros((n*m,1), int)  #dummy variable

    #compute the bitwise xor matrix
    M1 = bitxormatrix(genl1)
    M2 = np.triu(bitxormatrix(genl2),1)

    for i in range(m-1):
        for j in range(i+1, m):
            [r,c] = np.where(M2 == M1[i,j])
            for k in range(len(r)):
                VT[(i)*n + r[k]] = 1;
                VT[(i)*n + c[k]] = 1;
                VT[(j)*n + r[k]] = 1;
                VT[(j)*n + c[k]] = 1;

```

```
22                  if M is None:
23                      M = np.copy(VT)
24                  else:
25                      M = np.concatenate((M, VT), 1)
26
27                  VT = np.zeros((n*m,1), int)
28
29      return M
```

Questo è un elenco puntato

- primo elemento
  - primo sottoelemento
- secondo elemento
- terzo elemento

Questo è un elenco numerato

1. primo elemento
   (a) primo sottoelemento
2. secondo elemento
3. terzo elemento

Questa è una frase che dice qualcosa, e deve essere supportata da una citazione. Se gli autori sono più di due, scriviamo che Garousi et al. hanno detto qualcosa [**?**]. Se gli autori sono due, scriviamo che Bruegge e Dutoit hanno detto cose [**?**]. Quest'altra frase dice qualcos'altro, ma non so in quale paper l'ho letta Citazione necessaria. Questa frase dice cose supportate da diverse pubblicazioni [**?**, **?**, **?**, **?**, **?**, **?**, **?**]. In questa parte devo ricordarmi qualcosa NB ►***devo ricordarmi di questa cosa***◄ .

Questa parte del documento è stata modificata dopo che il relatore mi ha dato suggerimenti.

Invece qui ho avuto dei feedback dal relatore, e devo ricordarmi di applicare i commenti. aggiungi un paragrafo sulle research question

◎ **Our Goal.** Questo è l'obiettivo della tesi.

**RQ$_1$.** *Quale è la nostra prima Research Question?*

**Answer to RQ$_1$.** Questa è la risposta alla prima RQ.

**Finding 1.** Questo è il primo finding che riportiamo tra i risultati.

**Take-away Message #1.** Primo takeaway message.

CHAPTER 3

# Conclusions and Future work

# Bibliography

[1] "Iso/iec/ieee international standard - systems and software engineering – software life cycle processes," *ISO/IEC/IEEE 12207:2026(en)*, pp. 1–154, April 2026. (Cited on page 1)

[2] T. McKenzie, M. E. Trujillo, and S. Hoermann, "Software engineering practices and methods in the game development industry," 10 2019, pp. 181–193. (Cited on page 1)

[3] M. Bengtsson, T. Wilson, P. Zackariasson, and T. L. Wilson, "Paradigm shifts in the video game industry," *Competitiveness Review*, vol. 20, no. 2, pp. 139–151, 03 2010. [Online]. Available: https://doi.org/10.1108/10595421011029857 (Cited on page 2)

[4] D. Lin, C.-P. Bezemer, and A. E. Hassan, "An empirical study of early access games on the steam platform," *Empirical Software Engineering*, vol. 23, 04 2018. (Cited on page 2)

[5] U. Rajapakshe, "Development centric player feedback analysis for video games: A review," in *2019 International Conference on High Performance Big Data and Intelligent Systems (HPBD&IS)*, May 2019, pp. 190–194. (Cited on page 2)

[6] A. Al Tawil, H. Fathi, S. Alzubi, A. Shaban, and L. Almazaydeh, "Advanced feature extraction and machine learning techniques for classifying steam game

feedback," *International Journal of Interactive Mobile Technologies (iJIM)*, vol. 19, pp. 107–125, 01 2025. (Cited on page 2)

[7] Z. Ji, N. Lee, R. Frieske, T. Yu, D. Su, Y. Xu, E. Ishii, Y. J. Bang, A. Madotto, and P. Fung, "Survey of hallucination in natural language generation," *ACM Computing Surveys*, vol. 55, no. 12, p. 1–38, Mar. 2023. [Online]. Available: http://dx.doi.org/10.1145/3571730 (Cited on page 3)

[8] Y. Li and K. Aslam, "Ethical issues in video games: Insights from reddit discussions," in *2025 IEEE/ACM 47th International Conference on Software Engineering: Software Engineering in Society (ICSE-SEIS)*. IEEE Press, 2025, p. 185–196. [Online]. Available: https://doi.org/10.1109/ICSE-SEIS66351.2025.00024 (Cited on page 3)

# Acknowledgements

Acknowledgements here...

*This thesis has contributed to planting a tree in Kenya through the Treedom project.*

https://www.treedom.net/it/user/sesalab/event/sesa-random-forest